Face Recognition Using Machine Learning

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By

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CERTIFICATE

This is to certify that the term paper report titled "**Face Recognition Using Machine Learning**" submitted by **Naram Bharat Kumar** bearing **21341A1285** has been carried out in partial fulfillment for the award of B. Tech. degree in IT to **JNTU-GV** is a record of bonafide work carried out under our guidance and supervision. The report embodied in this paper has not been submitted to any other universities or institutions for the award of any degree or diploma.

Signature of the Supervisor	Signature of the H.O.D
Mr. M. Hari Krishna,	**Dr. V. Vasudha Rani**
Asst. Professor**,**	Associate Professor & HOD
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ABSTRACT

Most of the things in this world have become automated because of technological improvements. Face recognition is a popular application in machine learning that aims to identify and verify individuals based on their facial features. This paper explores an overview of key methodologies in face recognition by using machine learning. It involves extracting key facial attributes such as positions of the nose, eyes, mouth, and ear by using different algorithms we used to compare and match these features with a database of known faces. This process typically includes steps like face detection, feature extraction, and classification. Popular techniques used in face recognition include deep learning models like CNNs (Convolutional Neural Networks) and algorithms like PCA (Principal Component Analysis) and LDA (Linear Discriminant Analysis). Face recognition has various applications like FRS (Face Recognition System), and mobile phones and it includes security and personalized user experiences.

Keywords: *Face recognition, Deep learning, Machine learning, Computer neural networks, Principal Component Analysis, and Linear Discriminant Analysis.*

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1. INTRODUCTION

Face recognition using machine learning is a technology that uses algorithms to identify and verify people's faces. It involves training a model with a large dataset of labelled faces and then using that model to recognize and match faces in real-time. Face Recognition is a category of biometric software that maps individual facial features and stores the data as a face print. Both Machine Learning and Image Processing are the backbones of Face Recognition. Convolution Neural Network in Face Recognition is the widely used algorithm. CNN is a type of deep-learning neural network consisting of five layers. Input, Convolution, Pooling, Fully Connected, and Output layer. The role of the Convolution layer is to apply some learnable filters to the original image to extract the feature from a given dataset. The pooling layer down samples the image to reduce the computation and the fully connected layer makes the final prediction.

In the era of rapid technological evolution, the automation of various facets of daily life has become ubiquitous, driven by continuous improvements in machine learning. Among the prominent applications, face recognition stands out as a transformative technology aimed at identifying and verifying individuals based on their unique facial features. This paper presents a comprehensive overview of key methodologies employed in face recognition using machine learning techniques. In essence, this paper serves as a comprehensive guide to the multifaceted realm of face recognition through machinelearning, offering valuable insights into the intricacies of algorithmic approaches, their applications, and the evolving landscape of this transformative technology.

2. LITERATURE SURVEY

The following research papers are studied in detail to understand the proposed recommendation technique and experimental result for predicting the output.

2.1. Pandey, Ishan Ratn, Mayank Raj, Kundan Kumar Sah, Tojo Mathew, and M. S. Padmini. "Face Recognition Using Machine Learning." Int. Res. J. Eng. Technol 6 (2019): 3772-377

- This paper proposes a developed CNN model by adding Batch Normalization.
- To obtain a powerful high accuracy recognition algorithm.
- Images of 4 people taken at different times with different lighting conditions taken between 2/03/2019 to 8/03/19.
- SoftMax Classifier is used to classify the faces in a fully connected layer step.
- It extracted the features from the trained CNN and feeding them as input to the Fully connected layer with softmax activation for classification.
- Accuracy of Convolutional Neural Networks is 98.4%

2.2 Ikromovich, H. O., & Mamatkulovich, B. B. (2023). FACIAL RECOGNITION USING TRANSFER LEARNING IN THE DEEP CNN. Open Access Repository, 4(3), 502-507

- This approach aims to improve the accuracy of Facial Recognition models using Transfer Learning.
- In this paper Fine-tuning pre-trained CNN models achieved higher accuracy than feature extraction with a linear classifier.
- Labeled Faces in the Wild (LFW) dataset contains over 13,000 face images of 5,749 individuals.
- Fine-tuning VGG-16 achieved higher accuracy than fine-tuning ResNet-50.
- Process of face recognition and classification involves various steps such as preprocessing, detection, orientation, extraction of features and classification of emotion.

2.3 Hussain, S. A., & Al Balushi, A. S. A. (2020). A real-time face emotion classification and recognition using a deep learning model. In Journal of Physics: Conference Series (Vol.1432, No. 1, p. 012087). IOP Publishing.

- This paper proposes and investigates the Softmax function which is used to obtain and classify the image between 0 and 1.
- The Kenya Dryland Education Fund data set is used.
- Facial recognition is built with the Convolutional Neural Network model where the facial features are extracted and classified as per VGG 16 model.
- The performance measures are validated with the CNN model designed with an accuracy of 88%.
- Process of face recognition and classification involves various steps such as preprocessing, detection, orientation, extraction of features and classification of emotion.

2.4 Singh, A., Bhatt, S., Nayak, V., & Shah, M. (2023). Automation of surveillance systems using deep learning and facial recognition. International Journal of System Assurance Engineering and Management, 14(Suppl 1), 236-245.

- This paper presents a system that can identify and classify individuals in surveillance footage.
- Markov Random Filed is used to train a deep generation model of face pictures.
- HOG (Histogram Orientation Gradient) face detector is used. It is a feature descriptorused in computer vision and image processing for object detection
- The VGGFace model has higher accuracy.
- Labelled Faces in the Wild(LFW) data set is used.It contains over 13,000 images from hundreds of different celebrities.

2.5 Sun, X., Wu, P., & Hoi, S. C. (2018). Face detection using deep learning: An improved faster RCNN approach. Neurocomputing, 299, 42-50

- In this paper Convolution Neural Network (CNN) and VGG16 models are used.
- Histogram Orientation Gradient (HOG), Scale Invariant Feature Transform(SIFT), Speed Up Robust Feature(SIFT), and Aggregate Channel Feature(ACF) face detectors are used.
- The author of this paper Proposed Recurrent Neural Network Architecture for Face Detection.
- A Wider Face data set is used.
- The proposed face recognition system is robust against various face presentation attacks, such as printed paper, video replay, and silicone masks.

3. METHODOLOGY

3.1 Face Recognition Using Machine Learning

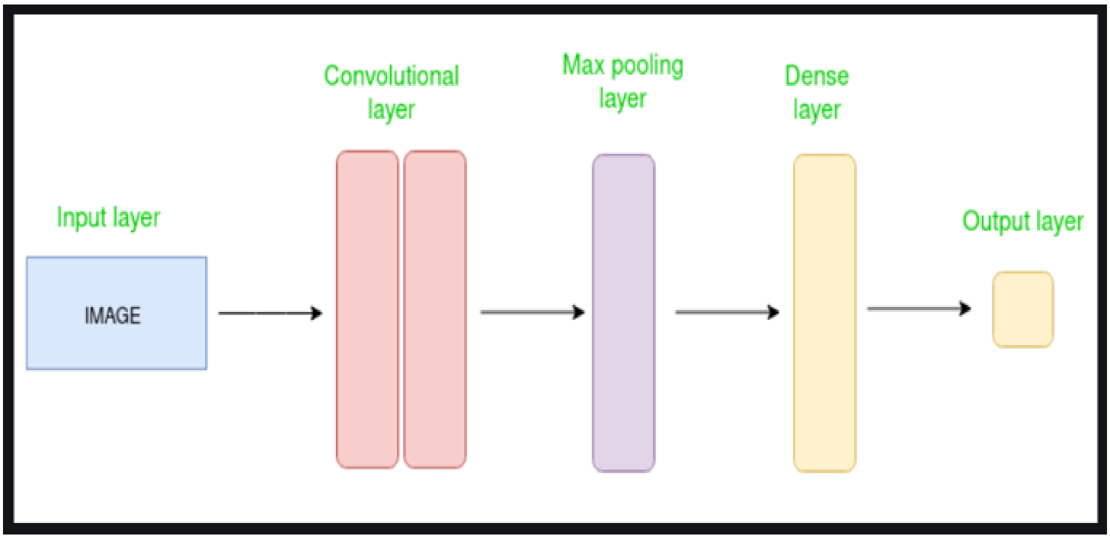


Fig1: Block Diagram of CNN

Layers of CNN:

- **The convolution layer** performs the core building block of a Convolutionary Network that performs most of the heavy-lifting computations. The convolution layer's main purpose is to extract features from the image-based input data. By learning image features using small squares of input image, Convolution preserves the spatial relationship between pixels.Using a set of learnable neurons, the input image is compressed. This creates a feature map in the output image and then feeds the feature maps to the next convolution layer as input data.
- **The pooling layer** reduces each activation map's dimensionality but still has the most important information. The images input are divided into a set of rectangles that are not overlapping. A non-linear operation such as average is used to down-sample each region. This layer achieves better generalization, faster convergence, robust translation and distortion, and is usually placed between layers of convolution.
- **The term Fully Connected Layer** (FCL) refers to each filter connected in the next layer in the previous layer. The output from the layers of convolution, pooling, and ReLU are incarnation of the input image's high-level features. The aim of using the FCL is to use these features to classify the input image into different classes based on the training dataset. FCL is considered to be the final layer of pooling feeding the features to a classifier using the activation function of Softmax. The sum of Fully Connected Layer's output probabilitiesis 1. Using the Softmax as the activation function ensures this. The Softmax function takes an arbitrary real-evaluated scores vector and transforms it into a value vector between 0 and1 that sums up to 1.
- **ReLU** is a non-linear operation that involves units that use the rectifier. It is an element- wise operation which means that it is applied per pixel, reconstituting all the negative values by zero in the feature map. To understand how ReLU operates, we assume that in theliterature for neural networks there is a neuron input given as x and from that the rectifier is defined as

$$f(x)= \max(0,x).$$

3.2 FACIAL RECOGNITION USING TRANSFER LEARNING IN THE DEEP CNN

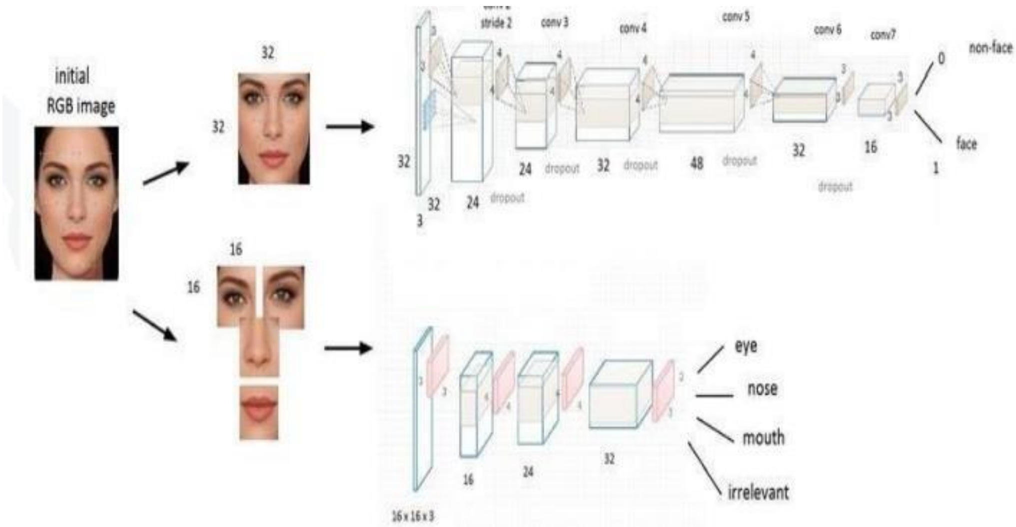


Fig: Pre-trained CNN Model for Facial Recognition

Example of a pre-trained CNN model used for facial recognition using transfer learning. The pre-trained CNN model is first loaded and the fully connected layers are replaced with new ones. The model is then fine-tuned using a small dataset of facial images to adapt it to the specific facial recognition task.

Labeled Faces in the Wild (LFW) dataset, which contains over 13,000 face images of 5,749 individuals. We used a deep CNN architecture that consists of convolutional layers, max-pooling layers, and fully connected layers. We compared the performance of our transfer learning approach with that of a baseline CNN that was trained from scratch on the same dataset. We conduct experiments on two facial recognition datasets: LFW (Labeled Faces in the Wild) and CelebA. LFW contains 13,000 images of 5,749 people, and CelebA contains 202,599 images of 10,177 celebrities. We use VGG-16 and ResNet-50 as pretrained CNN models for feature extraction and fine-tuning, respectively. We compare the performance of fine-tuning and feature extraction with a linear classifier, and we vary the number of training samples from 50 to 5,000.

3.3 A real-time face emotion classification and recognition using a deep learning Model

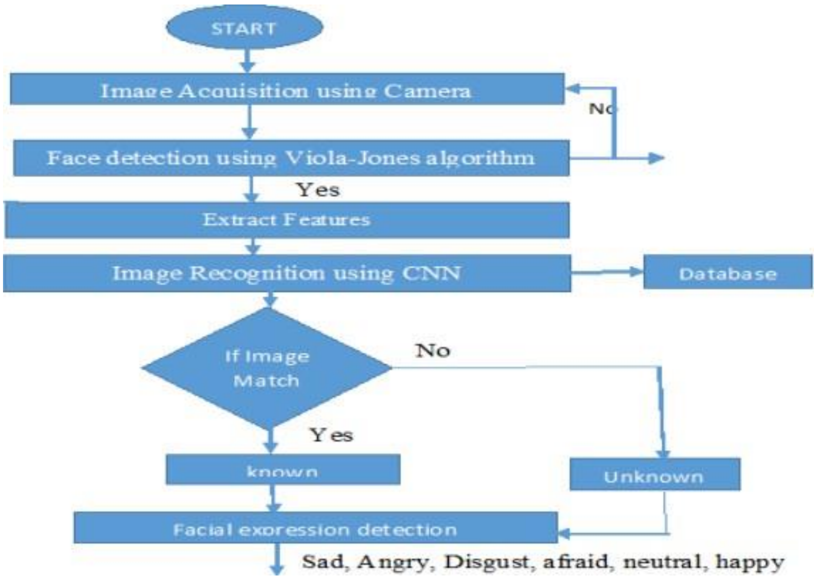


Fig: Face detection and recognition flow chart

The Real-time face detection, recognition, and classification is depicted in the figure. The captured image is bounded in a box and converted into a binary pattern to specify it as a feature vector and stored in a database. The images are trained to match the input image and also classify the expression of facial features as sad, angry, happy, disgusted, and neutral. In the training process, there are a total of seven steps which are represented below loading the dataset, preprocessing, augmenting the data as a feature vector, building and compiling the design model, training and storing the feature vector, and validating the test model.

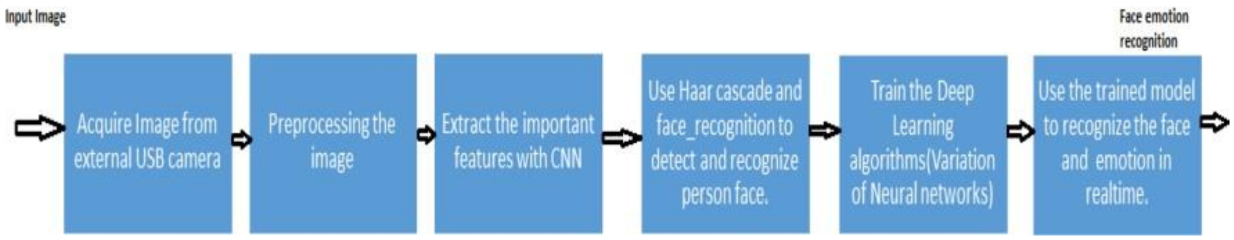
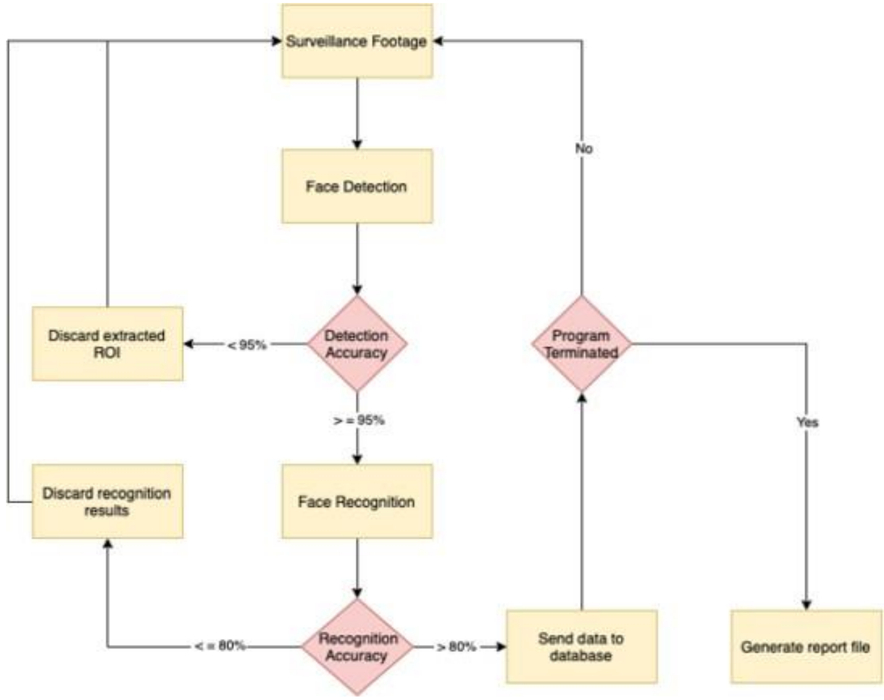


Fig: Block diagram of Face detection and recognition classification

The proposed system and its functioning are shown in the figure. The individual functional blocks have their specific operation. The image is captured using an HP 3000 camera which is preprocessed further to feed the neural networks. The captured image is modeled usingnetwork architecture to build and train the dataset for identification and emotion classification.

3.4 Automation of surveillance systems using deep learning and facial recognition



Face detection

In 2017, OpenCV 3.3 was released, which included a much improved DNN module that works with a variety of deep learning frameworks, including Cafe, TensorFlow, and Torch/PyTorch. Most crucially, regardless of where it was generated, the module could be imported and utilized across any framework. The key contributor to the DNN module, Aleksandr Rybnikov, also included a state-of-art face detector based on the Single ShotDetector (SSD) framework with a ResNet base network in the official release of OpenCV which can be found in the face_detector sub-directory of the DNN samples. With the pre- trained weights, faces could be detected in image stills, videos, video streams, and livewebcams using the DNN face detector. Moreover, due to its lightweight architecture, itperforms admirably in terms of operational speed, and CPU memory consumption stayscomparatively low suitable for real-time detection. The input image passed to the face detectoris first pre-processed using the DNN library which includes setting the image dimensions and normalization. The confidence is retrieved after each pass of the input image through the neural network, and weak detection is ruled out if the current confidence falls below the set threshold. The collected ROI is then sent through the face recognition pipeline for identification.

Face Recognition

On the internet, there are a plethora of face recognition techniques to choose from. Choosing the right technique for the job, on the other hand, is the most difficult part of the project. We began with a deep learning-based local neural network for extracting facial features and computing embeddings. After that, the embeddings were used as a training input for the SVM classifier. The trained model was stored locally and then deployed for recognition later. Although the performance was satisfactory when tested on two individuals, the accuracy dropped as the input dataset grew larger. In addition, we experimented with the local DNN by adding additional convolutional layers, including max pooling and dropout layers, using various optimizers, and altering the DNN's input image size. We even used a face aligner, and OpenCV's deep learning libraries for image optimization and pre-processing, and converted the input images from RGB to grayscale, and YCbCr. Unfortunately, the local DNN was notup to the task of real-time recognition, and the model would mistakenly identify two separate individuals wearing specs as the same person, and a single individual under various lighting circumstances as a different person.

3.5 Face detection using deep learning: An improved faster RCNN approach.

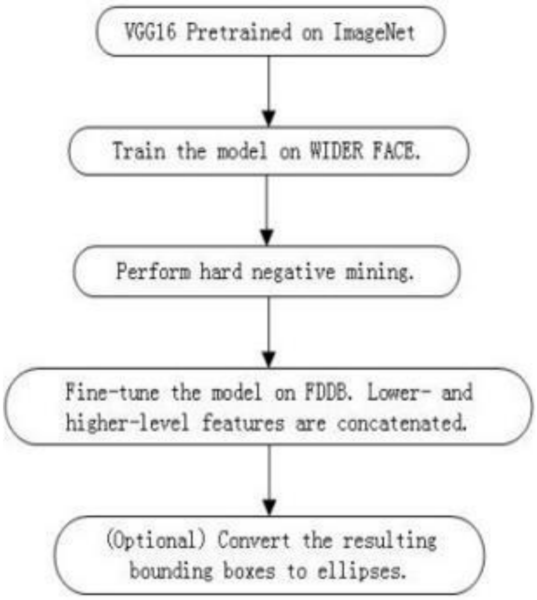


Fig: Flowchart of the training procedure of the proposed deep learning scheme

Our method follows the similar deep learning framework of Faster RCNN, which is a state-of-the-art deep learning scheme for generic object detection [10]. It essentially consists of two parts: (1) a Region Proposal Network (RPN) for generating a list of region proposals that likely contain objects, or called regions of interest (RoIs); and (2) a Fast RCNN network for classifying a region of image into objects (and background) and refining the boundaries of those regions. In this work, we propose to extend the Faster RCNN architecture for face detection and train our face detection model by following the proposed procedure as shown in Figure.

4. RESULTS AND DISCUSSIONS

S.No	Paper	Data set	Methods	Results
1.	*Pandey, Ishan Ratn, Mayank Raj, Kundan Kumar Sah, Tojo Mathew, and M. S. Padmini. "Face Recognition Using Machine Learning." Int. Res. J. Eng. Technol 6 (2019): 3772-377*	Images of 4 people taken between 2/03/19 to 8/03/19	Convolution Neural Network	Accuracy:98.4%
2.	Sun, X., Wu, P., & Hoi, S. C. (2018). Face detection using deep learning: An improved faster RCNN approach. *Neurocomputing*, *299*, 42-50.	Wider Face data	Convolution Neural Network	
3.	Ikromovich , H. O., & Mamatkulovich , B. B. (2023). FACIAL RECOGNITION USING TRANSFER LEARNING IN THE DEEP CNN. Open Access Repository, 4(3), 502-507.	Labeled Faces in the wild	Fine-tuning pre-trained CNN models	Fine-tuning VGG-16 achieved higher accuracy than fine-tuning ResNet-50.
4.	Hussain, S. A., & Al Balushi, A. S. A. (2020). A real-time face emotion classification and recognition using a deep learning model. In *Journal of Physics: Conference Series* (Vol. 1432, No. 1, p. 012087). IOP Publishing.	Kenya Dryland Education Fund	VGG-16	Accuracy:88.0%
5.	Singh, A., Bhatt, S., Nayak, V., & Shah, M. (2023). Automation of surveillance systems using deep learning and facial recognition. International Journal of System Assurance Engineering and Management, 14(Suppl 1), 236-245.	7500 live images of over 26 people, in various light conditions and poses	VGGFace model	Accuracy:96.0% .

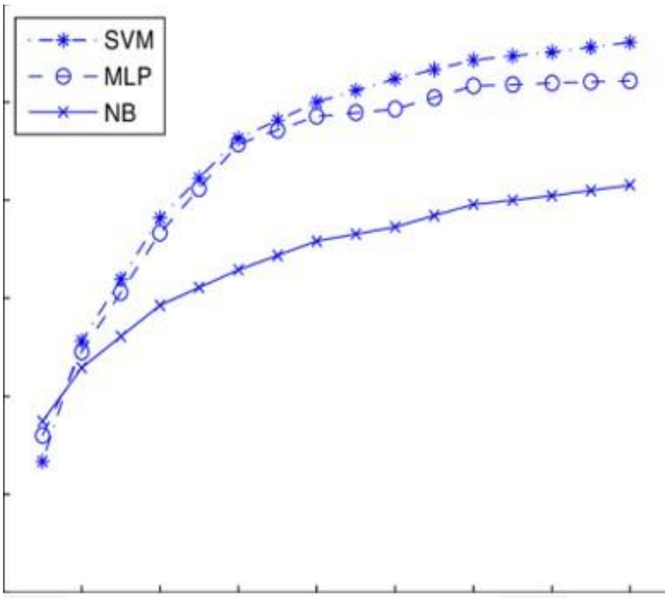
4.1 Face Recognition Using Machine Learning.

Our face database contains images of 4 people taken at different times with different lighting conditions taken between 2/03/2019 to 8/03/19. Each individual in the database is represented by a hundred colored JPEG images with cluttered backgrounds taken at 640 pixels resolution. In these images, the average size of the faces is 150bis150 pixels. The pictures show frontal and/or inclined faces with different conditions of lighting and scale.

CNNs are a category of neural networks that are highly effective in areas such as facerecognition and classification. CNNs are a type of feed-forward, multilayered neural network. CNNs consist of neurons with learning weights. Each filter takes certain inputs, converts and follows them with a non-linearity.

The prominent features of the proposed algorithm is that it uses batch normalization for the outputs of the first and final convolution layers and higher accuracy rates are achieved by the network. Softmax Classifier is used to classify the faces in a fully connected layer step. Our Face Database tested the performance of the proposed algorithm. The results showed satisfactory rates of recognition according to literature studies.

4.2 FACIAL RECOGNITION USING TRANSFER LEARNING IN THE DEEP CNN



Accuracy vs Number of Training Samples

Comparison of the accuracy of facial recognition models using different transfer learning methods with varying numbers of training samples. Fine-tuning and feature extraction achieved high accuracy even with a small number of training samples while training from scratch required a larger number of training samples to achieve comparable accuracy.

For fine-tuning, we freeze the first few layers of the pre-trained CNN model and finetune the remaining layers on the facial recognition task. We use a learning rate of and train the model for 20 epochs with a batch size of 32. For feature extraction, we extract the features from the pre-trained CNN model and train a linear classifier on the extracted features. We use logistic regression as the linear classifier and train it with a.

4.3 A real-time face emotion classification and recognition using deep learning Model

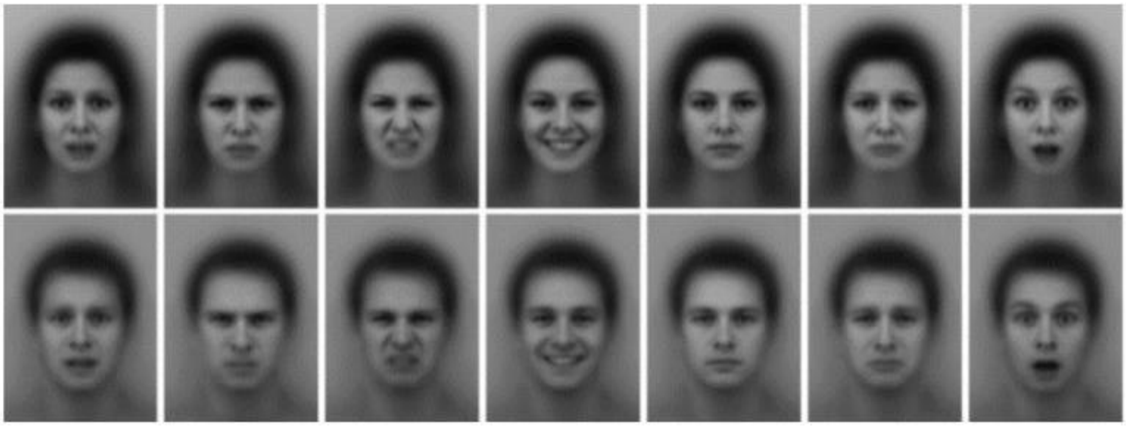


Fig: Sample KDEF data set (KDEF | The Kenya Dryland Education Fund 2019)

Building a dataset to classify emotion using a convolution neural network layer.

The Kenya Dryland Education Fund data set as in the figure has 4900 pictures whereas 1999 pictures were removed as they were in side postures and the dataset with 2901 pictures by splitting and training the dataset into 70 % and 30% respectively. Similarly to my images, my dataset was also created with 100 images.

The captured image is bounded in a box and converted into the binary pattern to specify it as a feature vector and stored in a database. The images are trained to match the input image and also classify the expression of facial features as sad, angry, happy, disgusted, and neutral. In the training process, there are seven steps represented below loading the dataset, preprocessing, augmenting the data as a feature vector, building and compiling the design model, training and storing the feature vector, and validating the test model.

4.4 Automation of surveillance systems using deep learning and facial recognition

With our original dataset, we evaluated numerous neural models, including VGG16, a 16-layer neural network that won the Imagenet competition in 2014, VGG19, which has 19 layers, VGGFace, which is based on VGG16, Inception v3 with 48 layers, and ResNet-50, which has 50 deep layers. Every model was re-trained by keeping the top layers and adding activation and dropout layers as appropriate. As a result of employing pre-trained weights and neural architecture, we were successful in improving the accuracy of our face recognition system.
The facial recognition model received live input from the surveillance camera. The model was trained to recognize 26 distinct individuals, and whenever one was identifed, an accuracy associated with the recognition was generated as well. Furthermore, we used XAMPP to set up a local Apache HTTP server with MySQL as the database management system. If an individual was identifed with a precision of over 80%, the name label and a unique timestamp were sent to the database. Our algorithm was designed to run for a specifc amount of time, hence, the last recorded entry of every unique individual identifed by the system was pulled from the database and saved in an excel fle on the local computer before the system was terminated.

4.5 Face detection using deep learning: An improved faster RCNN approach.

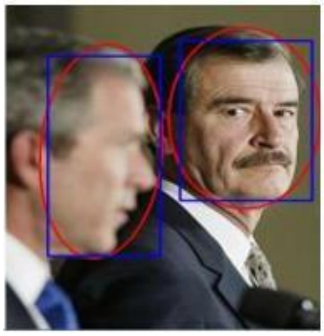


Fig:1

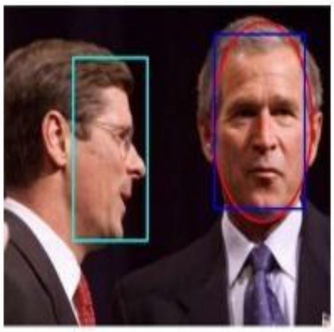


Fig:2

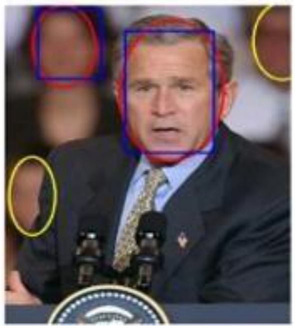


Fig:3

In addition to the quantitative evaluation results, we also randomly choose some qualitative results of face detection examples for different cases, as shown in Fig1, Fig2, and Fig3. For example, Fig 1 demonstrates that our model can detect some difficult cases, such as non- frontal faces, heavily occluded faces, faces with low resolution, and faces with extreme poses and/or illumination. Fig 2 lists some selected false positives, where it seems that most of the false positives are missing annotations. Fig 3 lists some of the false negatives, which include some very challenging cases, such as blur faces, heavily occluded faces, and extremely small faces.

In this work, we proposed a new method for face detection using deep learning techniques. Specifically, we extended the state-of-the-art Faster RCNN framework for generic object detection, and proposed several effective strategies for improving the Faster RCNN algorithm for resolving face detection tasks, including feature concatenation, multi-scale training, hard negative mining, and proper configuration of anchor sizes for RPN, etc. We conducted an extensive set of experiments on the well-known FDDB testbed for face detection benchmark, and achieved the state-of-the-art results which ranked the best among all the published methods. Future work will further address the efficiency and scalability of the proposed method for real-time face detection.

5. CONCLUSION

In conclusion, face recognition using machine learning operates by extracting features from facial images through a pre-trained model. These features are transformed into embeddings, allowing for efficient comparison with stored identities in a database, The system then decides the identity of the input face, providing a powerful tool for various applications. Some algorithms used in this paper are a Convolutional Neural Network for feature extraction, a Histogram of Orientation and Gradientation for extracting the features from an image, VGG16 can serve as a feature extractor by using its convolutional layers to extract relevant facial features from images, Multi-task Cascaded Neural Network is used to locate and align the facial landmarks (such as eyes, nose, and mouth) and. However, the success of the system depends on the training data.

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